

HALSUA KANALA PUROLA, Finland

WMO: 027250

Lat: 63.446N

Lon: 24.444E

Elev: 502

StdP: 14.43

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.3	-10.9	-22.1	1.7	-17.3	-15.2	2.5	-10.9	13.5	29.1	12.2	28.9	0.7	120	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	80.4	64.8	76.2	62.5	72.8	60.9	67.2	75.8	64.8	73.2	62.8	70.2	5.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	91.0	70.4	61.9	84.2	68.2	59.5	77.3	65.8	31.9	75.5	30.1	73.3	28.6	70.0	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.3	9.4	DB	-19.4	83.2	8.6	4.9	-25.6	86.7	-30.7	89.6	-35.5	92.3	-41.8	95.9	(4)
(5)			WB	-19.7	69.2	8.5	3.6	-25.8	71.7	-30.8	73.8	-35.6	75.8	-41.8	78.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	15.0	18.5	25.3	35.6	47.1	53.9	59.7	56.4	48.4	37.1	30.4	23.2
	DBStd	17.17	12.95	12.92	9.58	6.90	8.12	6.86	6.34	5.95	6.24	6.91	8.19	11.11
	HDD50	5296	1086	881	765	433	153	35	4	13	103	403	589	831
	HDD65	10020	1551	1301	1230	881	557	339	187	274	499	866	1039	1296
	CDD50	787	0	0	0	1	62	152	306	211	53	2	0	0
	CDD65	38	0	0	0	0	1	6	24	7	0	0	0	0
	CDH74	584	0	0	0	0	57	108	339	80	0	0	0	0
	CDH80	103	0	0	0	0	3	16	73	11	0	0	0	0
Wind	WSAvg	4.3	3.9	4.5	4.8	4.3	4.6	4.3	3.6	3.8	4.0	4.5	4.5	4.4
Precipitation	PrecAvg	24.4	1.7	1.4	1.3	1.3	1.9	2.6	3.2	2.8	2.2	2.4	2.1	2.0
	PrecMax	32.2	3.5	2.7	2.2	2.9	3.7	4.5	6.1	6.3	5.3	4.4	3.9	4.1
	PrecMin	18.5	0.5	0.2	0.2	0.3	0.3	0.8	0.9	1.1	0.7	0.6	0.2	0.7
	PrecStd	3.7	0.7	0.6	0.5	0.7	0.9	1.1	1.1	1.3	1.3	1.1	0.9	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.4	41.2	46.9	64.9	79.8	82.3	85.5	81.4	68.4	56.9	49.1	41.3
		MCWB	36.3	37.7	38.8	49.7	59.2	63.4	67.0	67.8	59.4	53.2	47.4	40.2
	2%	DB	34.7	38.1	43.0	57.8	74.8	76.9	82.1	75.4	64.8	52.0	44.9	37.8
		MCWB	33.7	35.4	37.3	46.7	57.6	62.6	66.9	63.0	57.5	49.8	43.6	36.6
	5%	DB	32.9	35.2	39.5	51.8	70.1	72.5	77.6	71.7	61.7	49.7	42.7	35.8
		MCWB	32.2	33.8	35.3	43.0	55.5	59.5	64.6	61.0	55.2	47.8	42.0	34.9
	10%	DB	31.5	33.2	36.9	48.0	64.5	68.6	73.9	68.2	58.8	47.1	40.2	34.2
		MCWB	30.9	32.2	34.0	41.0	51.8	58.1	62.9	59.9	54.5	45.5	39.6	33.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.5	37.8	40.3	51.1	62.0	66.6	71.0	69.5	60.9	53.5	47.8	40.7
		MCDB	37.1	40.0	44.6	60.3	74.8	75.3	80.5	79.1	65.8	55.5	48.8	41.5
	2%	WB	33.8	35.7	37.8	47.7	59.1	64.1	68.3	65.3	58.6	50.7	43.9	36.8
		MCDB	34.2	37.5	41.4	56.6	70.6	73.5	77.0	72.3	62.8	51.8	44.7	37.5
	5%	WB	32.5	34.1	36.1	44.2	56.6	61.6	66.1	62.9	56.7	48.0	41.9	34.9
		MCDB	33.0	35.2	38.9	50.2	68.6	70.7	74.9	68.3	60.1	49.2	42.5	35.4
	10%	WB	30.9	32.6	34.2	41.3	53.3	58.9	64.2	61.0	55.0	45.5	39.7	33.6
		MCDB	31.5	33.4	36.4	46.9	64.1	66.5	72.5	66.3	58.0	46.9	40.3	34.1
Mean Daily Temperature Range	5% DB	MDBR	11.4	12.8	15.8	17.6	22.1	20.3	20.2	18.7	15.6	10.8	6.9	8.2
		MCDBR	6.8	9.2	14.4	26.2	31.9	27.9	26.8	25.0	21.1	12.9	8.5	7.7
		MCWBR	6.7	7.8	10.3	16.2	17.0	14.9	13.6	14.0	14.1	10.6	8.1	7.4
	5% WB	MCDBR	6.9	9.0	13.1	24.0	29.1	23.7	23.9	20.2	17.3	11.6	8.1	7.5
		MCWBR	6.8	8.0	10.0	15.6	16.0	13.3	12.9	11.9	12.2	10.1	7.8	7.3
Clear-Sky Solar Irradiance	taub	0.221	0.282	0.309	0.350	0.345	0.346	0.367	0.352	0.328	0.308	0.374	0.356	
	taud	2.006	2.144	2.209	2.332	2.449	2.466	2.423	2.460	2.516	2.389	2.701	2.638	
	Ebn at Noon	122	202	247	263	276	278	267	261	246	197	113	67	
	Edh at Noon	10	20	28	31	30	31	31	28	22	16	9	5	
All-Sky Solar Radiation	RadAvg	53	231	705	1184	1539	1676	1578	1201	688	266	66	23	
	RadStd	6	42	68	127	176	148	162	128	76	51	13	3	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		+0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page